Topic 2.4 Unification and Resolution in FOL

a) Definition

- Unification is a process of making sentences equal using substitution.
- A unifier is a substitution that makes sentences equal.
- Most general unifier (mgu) is one, which may have some instances, but which is not an instance of any other.

b) Basic idea of the process

- Different predicates or constants do not match.
- A variable matches a term if the term does not contain the variable itself.

c) Simplified UNIFY algorithm

We assume that

- Two same-predicate atomic sentences are given in a set, S;

- The algorithm returns the mgu, θ;

- The algorithm can be used repeatedly for a set with more than 2 such sentences in a chain.

Major steps:

1) Set $\theta = \{\}$.

2) If SUBST(θ, S) contains only one element, then stop; θ is an mgu of S. Otherwise, find the first <u>disagreeing pair</u> of SUBST(θ, S).

3) If the disagreeing pair contains a variable v and a term t such that v does not occur in t (that is, t is another variable / a constant / a function without v), then obtain

   <u>$\theta = \theta \cup \{v / t\}$</u>, and go to step 2. Otherwise stop; S is not unifiable.

Example:

S = {P1("Km", x, y), P1(z, u, F1(z))}

1. θ = {}, SUBST(θ, S) = S.

2. θ = { z / "Km" },
   SUBST(θ, S) =
   {P1("Km", x, y), P1("Km", u, F1("Km"))}.

3. θ = { z / "Km", x / u },
   SUBST(θ, S) =
   {P1("Km", u, y), P1("Km", u, F1("Km"))}.

4. θ = {z / "Km", x / u , y / F1("Km") },
   SUBST(θ, S) = {P1("Km", u, F1("Km"))}.
   Terminate.

d) The <u>Resolution rule</u> of FOL

Example:

$P1(x) \vee P5(\text{"Km"}, x) \vee P3(y)$

$\neg P4(\text{"Sm"}, z) \vee \neg P5(\text{"Km"}, \text{"Rm"}) \vee P6(\text{"Hm"})$

$P1(\text{"Rm"}) \vee P3(y) \vee \neg P4(\text{"Sm"}, z) \vee P6(\text{"Hm"})$

The **rule**:

From two clauses,

$p_1 \vee p_2 \vee \dots \vee p_i$ and

$q_1 \vee q_2 \vee \dots \vee q_j$

we resolve

$\mathrm{SUBST}(\theta, p_1 \vee p_2 \vee \dots \vee p_{k-1} \vee p_{k+1} \vee \dots \vee p_i \vee q_1 \vee q_2 \vee \dots \vee q_{l-1} \vee q_{l+1} \vee \dots \vee q_j),$

where

p_x, q_y are FOL literals (positive or negative),

θ is a substitution and $\theta = \mathrm{UNIFY}(p_k, \neg q_l)$.